2016 EXAMINATIONS

Lancaster University

Part I (First Year)

SCHOOL OF COMPUTING AND COMMUNICATIONS

SCC.130 Information Systems (1 hour & 30 Minutes)

- **Answer any THREE out of the four questions.**
- **Use a separate answer book for each question.**

Question 1

1.a Define the principle of Moore's Law. What are the consequences of this law on the cost of data communications and storage? Using an example, explain how this law can benefit information systems.

[5 marks]

1.b Consider the five-component framework of an information system

i. Draw a representation of this framework. Label and briefly define the functionality of each component. **[5 marks]**

ii. Using an information system as an example, order the components in terms of increasing degree of difficulty of change and amount of disruption. Give reasons for your choice of order and how a change can affect the overall system. **[5 marks]**

iii. Identify two systems where the first creates more work for people and the second creates less work. Explain how the two systems differ and the implications this has for an organisation. **[4 marks]**

1.c An e-business application over the Web is based on the 3-tier architecture. Identify each tier and describe its function.

[6 marks]

[Total 25 Marks]

Question 2

In this question, we use the following three relations from a relational database.

Movies			
id	**title**	**released**	**Runtime**
2	Back to the Future Part II	1989-11-20	NULL
4	Twelve Monkeys	1995-12-27	129
5	Ocean's Eleven	2001-12-06	116
6	Ocean's Twelve	2004-12-09	125
7	Sin City	2005-03-31	124

actors		
id	**name**	**nationality**
1	Michael J. Fox	Canadian
2	Bruce Willis	NULL
3	Brad Pitt	American
4	Elijah Wood	American
5	James McAvoy	British
6	Clive Owen	British

acts_in		
a_id	**m_id**	**part**
1	2	Marty McFly
2	4	James Cole
2	6	NULL
2	7	Hartigan
3	5	Rusty Ryan
3	6	Rusty Ryan
3	4	Jeffrey Goines
4	2	Video Game Boy
4	7	Kevin
6	7	Dwight

2.a i. What properties must a candidate key possess? List the candidate keys for the actors relation.

[3 marks]

ii. What is the purpose of a primary key? What type of relational data integrity does it guarantee? List the primary keys required for each of the above relations.

[4 marks]

iii. What is the purpose of a foreign key? What type of relational data integrity does it enforce? List the foreign keys required for each of the above relations.

[4 marks]

Question 2 continues on the next page...

Question 2 continued.

2.b Write SQL statements that will generate answers to the following.

i. Write a statement for a scheme that would define the movies relation. Ensure that a value for the title attribute will be provided for every tuple.

[4 marks]

ii. Find the title of all movies with a runtime of 125 minutes or less.

[2 marks]

iii. List all movies that Elijah Wood has acted in, alongside the part played. List your results chronologically, earliest release date first.

[4 marks]

iv. List every actor in the actors table alongside the total number of movies they have acted in.

[4 marks]

[Total 25 Marks]

Question 3

3.a Define the legislation that protects the copyright of software in the United Kingdom. List two aspects that the legislation covers and two acts that are restricted under the legislation.

[5 marks]

3.b Give two causes of digital exclusion, either in the UK or overseas, and explain how such causes have affected people.

[6 marks]

3.c In the 1970s, a well-known Tobacco company commissioned a research study into whether there was a causal link between smoking cigarettes and lung cancer. However after seeing the results, the company decided not to publish the study's report. Explain what type of censorship this is and why it is this type, and explain the reasons for such censorship taking place.

[4 marks]

3.d Choose three malevolent online practices: define each and explain how these practices can impact users of the web.

[6 marks]

3.e In which contexts could a software developer's negligence have a significant effect? What steps could he/she take in order to ensure that he is not liable for negligence?

[4 marks]

[Total 25 Marks]

Question 4

4a. With the aid of examples, explain the rationale for adopting a structured approach to system analysis and design.

i. Compare the use of the total system approach to the piecemeal approach.

[2 marks]

ii. Give an example of a piecemeal approach and outline its limitations.

[3 marks]

iii. Present an example of a structured approach taking into account the building of the logical model. What benefits can this approach provide over the piecemeal approach?

[4 marks]

4.b Technology has been advancing rapidly, nowhere more effectively than in retail with the nature of shopping becoming increasingly Internet-focused. In particular, the popularity of mobile devices among consumers has encouraged high street retailers to develop e-business solutions for mobile systems. Suggest how Web. 2.0 technologies could be utilized to support e-commerce applications using the following technologies and applications:

i. Cloud computing

[4 marks]

ii. Rich internet applications

[4 marks]

Question 4 continues on next page….

Question 4 continued.

4.c Consider the two systems shown below in Figure 4.c.

i. What does each system represent?

[2 marks]

ii. How does the system on the left differ from the one on the right?

[3 marks]

iii. State any advantages and disadvantages associated with each system.

[3 marks]

[Total 25 Marks]

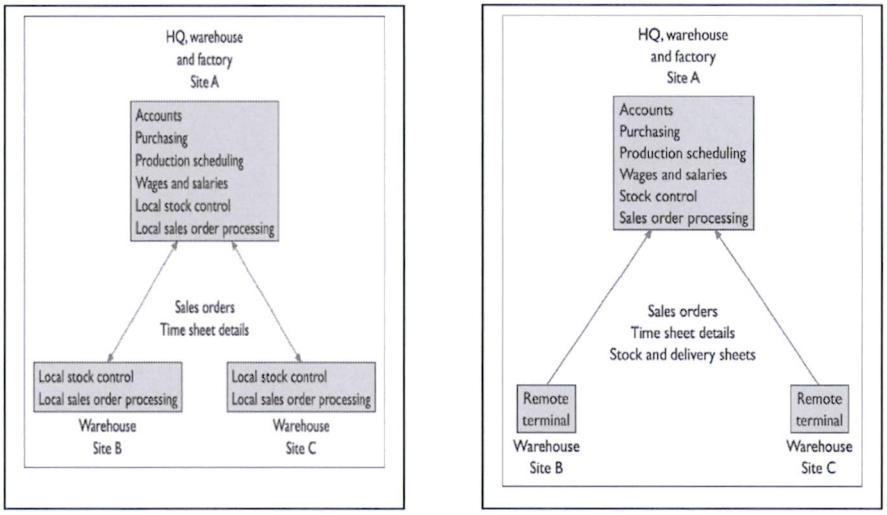

Figure for question 4.c

END OF PAPER